

- 2 The kinetic energy E_k of a rotating solid sphere is given by the following expression

$$E_k = \frac{2\pi^2 I}{T^2}$$

where I is the moment of inertia and T is the period of rotation of the solid sphere.

What is the unit for the moment of inertia I , expressed in SI base units?

- A $\text{kg m}^2 \text{s}^{-1}$
- B kg m^2
- C J s^2
- D N m s^2

Ans: B

$$E_k = \frac{2\pi^2 I}{T^2}$$

$$I = \frac{E_k T^2}{2\pi^2}$$

$$\begin{aligned}\text{unit for } I &= (\text{unit for } E_k)(\text{unit for } T)^2 \\ &= \{\text{unit for } (mv^2)\}(\text{unit for } T)^2 \\ &= (\text{kg})(\text{m}^2 \text{s}^{-2})(\text{s}^2) \\ &= \text{kg m}^2\end{aligned}$$

2 A ruler is supported horizontally by two pivots, as shown